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SAP PROJECT REPORT

Procure-to-Pay (P2P) System

Full Purchasing Cycle using SAP MM Module

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Abstract

The Procure-to-Pay (P2P) process is the backbone of procurement operations in any manufacturing or trading organization. It covers the complete journey from identifying a material requirement to making the final payment to the vendor. This project implements and documents a full end-to-end P2P simulation using SAP MM (Materials Management) for a fictitious company — Tech Edge Manufacturing Pvt. Ltd. The simulation covers every major SAP transaction: ME51N (Purchase Requisition), ME54N (PR Approval), ME21N (Purchase Order), MIGO (Goods Receipt — Movement Type 101), MIRO (Logistics Invoice Verification with 3-Way Match), and F-53 (Vendor Payment).

The system is built as an interactive web-based simulation that replicates SAP MM screens and workflows. It demonstrates core SAP concepts including release strategy, GR/IR clearing account logic, automated three-way matching, and complete document chain traceability. The project also covers the full organizational structure setup (Company Code, Plant, Purchasing Organization) and master data configuration (Vendor Master, Material Master, Info Records, GL Accounts) required before any SAP procurement transaction can be executed.

This report documents the complete process step by step, with screenshots taken directly from the simulation model, detailed explanations of accounting entries at each stage, unique learning outcomes, and planned future enhancements.

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1. Project Title

Procure-to-Pay (P2P) — Full Purchasing Cycle using SAP MM

This project implements a complete simulation of the SAP MM Procure-to-Pay business process for Tech Edge Manufacturing Pvt. Ltd., a fictitious manufacturing company set up specifically for this exercise. The simulation replicates the exact SAP workflow across all six core transaction codes — from Purchase Requisition creation through to final vendor payment — and documents the complete organizational setup, master data configuration, and accounting integration required for a fully functional SAP MM procurement environment.

2. Problem Statement

In most mid-to-large organizations, the procurement process is fragmented across multiple departments — Purchase, Finance, Stores, and Vendors — leading to delays, manual errors, duplicate orders, and poor spend visibility. Without an integrated system, purchase requisitions are raised informally, approvals happen via email, goods receipts are recorded in spreadsheets, and invoice matching is done manually. This results in:

- Delayed procurement leading to production stoppages
- Overspending due to lack of budget control mechanisms
- Vendor disputes caused by invoice mismatches
- Audit risks from incomplete, non-traceable approval chains
- No real-time inventory visibility or demand forecasting

This project demonstrates how SAP MM's Procure-to-Pay cycle solves these challenges by providing a fully integrated, automated, and auditable procurement workflow for a fictitious company — Tech Edge Manufacturing Pvt. Ltd.

The business impact of these problems is significant. Organizations without an integrated ERP procurement system typically experience a 40–60% longer purchase-to-delivery cycle, a 3–5% rate of duplicate or erroneous payments, and limited ability to enforce spending policies or generate accurate financial accruals at month-end. SAP MM's P2P cycle addresses each of these gaps through structured workflows, automated document generation, three-way matching, and real-time financial integration with SAP FI.

3. Solution & Features

The Procure-to-Pay cycle implemented in this project covers the complete SAP MM procurement workflow — from raising a Purchase Requisition to clearing the vendor invoice — inside a fictitious company setup with all required organizational structures.

3.1 Organizational Structure

In SAP, every procurement transaction must be anchored to a fully configured organizational hierarchy. The following entities were created for Tech Edge Manufacturing Pvt. Ltd. before any P2P transaction could be executed:

SAP Entity	Configuration Value	Description
Company Code	TE01	Legal entity for financial accounting
Plant	TE10	Main manufacturing plant
Storage Location	TW01	Warehouse / stores location
Purchasing Organization	TEPO	Central purchasing authority
Purchasing Group	T01	Team responsible for buying
Chart of Accounts	TECA	GL account structure

The Company Code (TE01) is the smallest organizational unit for which a complete set of accounts can be drawn up for external reporting. It is linked to the Chart of Accounts (TECA) which defines all GL accounts. The Plant (TE10) is linked to the Company Code and represents the physical production or storage facility. The Purchasing Organization (TEPO) is responsible for negotiating purchasing conditions with vendors and is assigned to both the Company Code and the Plant.

3.2 P2P Process Flow — Step by Step

The P2P cycle in SAP MM follows a strict document chain where each step references and is linked to the previous one. This linkage is what enables complete traceability from the initial need identification to final payment:

Step	T-Code	Process	Department
1	ME51N	Create Purchase Requisition (PR)	Requesting Dept.
2	ME54N	Approve Purchase Requisition	Manager
3	ME21N	Create Purchase Order (PO) from PR	Purchase Dept.

Step	T-Code	Process	Department
4	MIGO	Goods Receipt against PO (Movement Type 101)	Stores
5	MIRO	Logistics Invoice Verification (3-Way Match)	Finance
6	F-53	Vendor Payment Processing	Accounts Payable

3.3 Key Features

- Purchase Requisition creation and workflow approval (ME51N / ME54N)
- Automatic conversion of approved PR to Purchase Order (ME21N)
- Three-way matching: PO vs Goods Receipt vs Invoice (MIRO)
- Material Document and Accounting Document auto-generation on GR
- Stock update in MM (inventory) and FI (GR/IR clearing account)
- Vendor master data setup with payment terms (NET30)
- Info Records for material-vendor pricing relationships (ME11)
- Release Strategy for PR/PO approval levels
- Complete audit trail with document chain traceability

3.4 Master Data Configuration

Master data in SAP MM is the foundation on which all procurement transactions run. The following records were configured before executing the P2P cycle:

Master Data	T-Code	Key Details
Vendor Master	XK01	Tech Supply Co. (V001) — Payment Terms: NET30, Reconciliation Acc: 16000001
Material Master	MM01	Steel Rods (ROH), Copper Wire (ROH) — Basic Data, Purchasing, MRP, Accounting views
Purchase Info Record	ME11	Vendor-material price link — Rs.800/KG for Steel Rods from TechSupply Co.
Source List	ME01	Approved vendor list per material — prevents unauthorized sourcing
GL Accounts	FS00	Stock (40010001), GR/IR (29000001), Vendor Payables (16000001), Bank (11300001)

The Purchase Info Record (ME11) is particularly important — it stores the agreed price between a vendor and a material, which the system automatically proposes when a Purchase Order is created.

This eliminates manual price entry errors and ensures POs always reflect negotiated terms. The Source List (ME01) further controls procurement by restricting which vendors are eligible to supply a given material, preventing maverick buying.

4. Screenshots — P2P System in Action

The following screenshots are taken directly from the P2P simulation model built for this project. Each screenshot corresponds to a specific T-code step in the SAP MM Procure-to-Pay cycle with all relevant document numbers, field values, and system responses visible.

4.1 P2P Dashboard — System Overview (ME2N / Overview)

The dashboard displays real-time KPIs including Open PRs, Open POs, Goods Receipts for the month, and pending invoices. The P2P flow diagram at the centre highlights the current active step in the cycle.

The dashboard serves as the central monitor for the entire P2P process. The four KPI tiles at the top show Open PRs (3 — with 2 awaiting approval), Open POs (2 — including 1 partially received), GR This Month (5 — all matched), and Pending Invoices (2 — Rs.1,40,000 pending). The P2P flow diagram in the middle shows Steps 1 through 3 completed (green), Step 4 (MIGO — Goods Receipt) as the current active step, and Steps 5 and 6 pending. The Recent Activity table at the bottom provides a transaction log with document numbers, materials, vendors, amounts, and statuses.

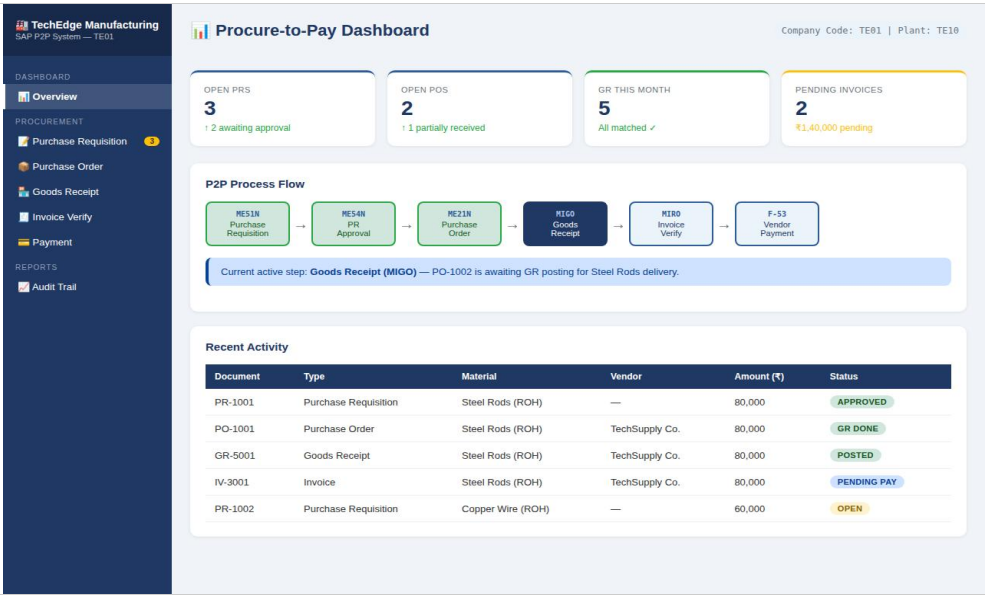

Fig 1: P2P Dashboard — TechEdge Manufacturing Pvt. Ltd. (Company Code: TE01)

4.2 Purchase Requisition — Create (ME51N)

The Purchase Requisition form captures header information (Document Type, PR Date, Plant) and line item details (Material, Quantity, Required Date, Estimated Price, Cost Centre, GL Account). The requesting department initiates procurement through this screen.

The ME51N screen is divided into a Header section and a Line Item section. The Header captures the Document Type (NB — Standard PR), PR Date, Requested By (RAVI.SHARMA), and Plant (TE10 — Tech Edge Main Plant). The Line Item section captures the Material Number (which auto-populates the description from the Material Master), Storage Location (TW01 — Main Warehouse), Quantity Required, Unit of Measure (KG), Required Date, Estimated Price per unit, Cost Centre (CC-MFG-001 — Manufacturing), GL Account (40010001 — Raw Material Purchases auto-populated), and a Reason / Justification field. The PR does not go to the vendor — it is an internal request document that triggers the approval workflow.

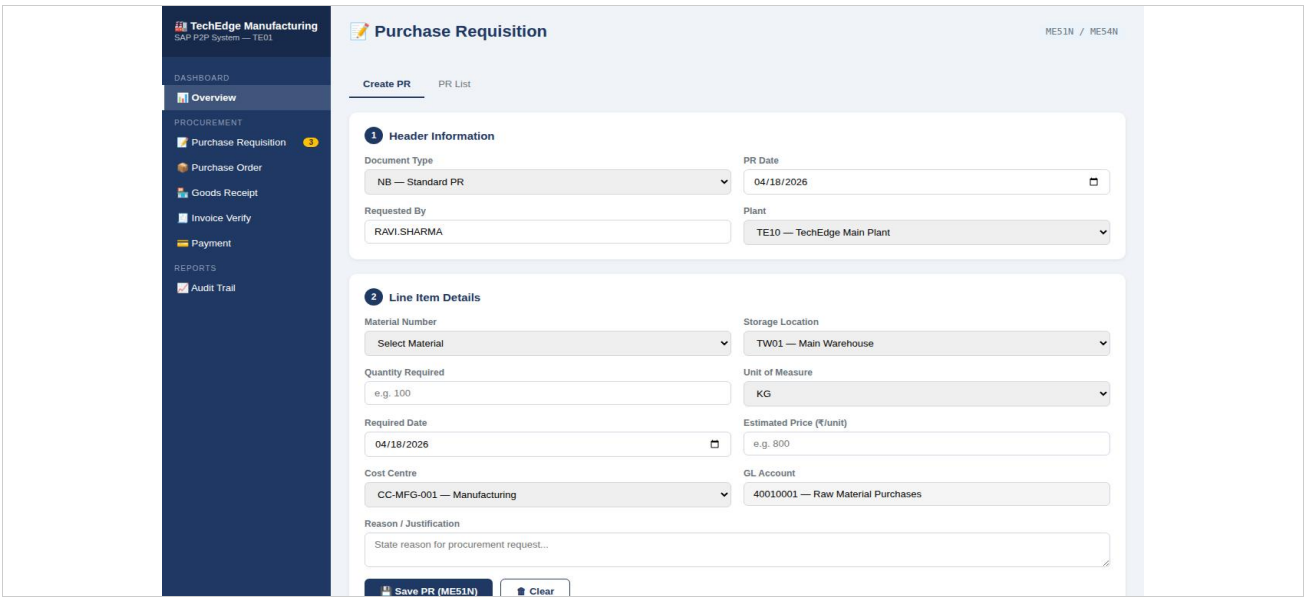

Fig 2: Create Purchase Requisition — ME51N (NB Standard PR, Plant TE10)

4.3 PR List & Approval (ME54N)

The PR List view shows all requisitions with their approval status. PRs pending manager approval are displayed with an Approve button. Once approved, the Release Indicator is set and the PR becomes eligible for Purchase Order conversion.

ME54N is the transaction used by managers to release (approve) Purchase Requisitions. The PR list shows PR-1001 (Steel Rods — Rs.80,000, Approved), PR-1002 (Copper Wire — Rs.60,000, Pending), and PR-1003 (Plastic Granules — Rs.40,000, Pending). The Release Strategy configured for this company requires manager approval for all PRs exceeding Rs.50,000. On clicking Approve, the system sets the Release Indicator on the PR, marking it as fully released and available for conversion to a Purchase Order in ME21N. Without this release, ME21N will not allow referencing the PR.

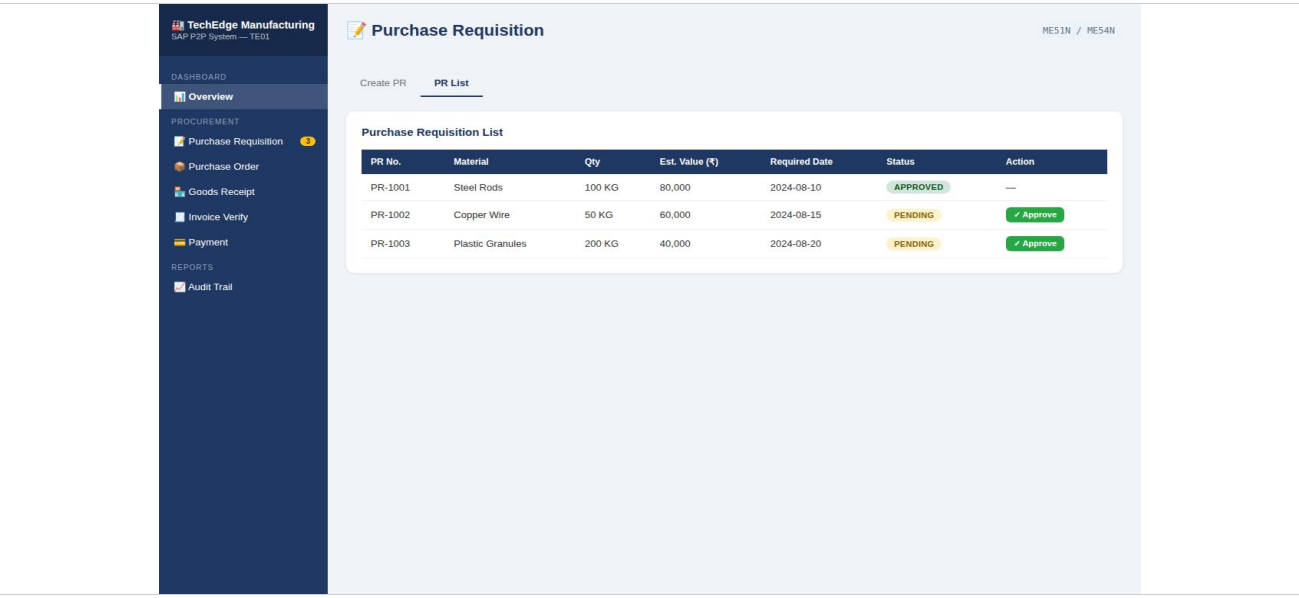

Fig 3: Purchase Requisition List & Approval Workflow — ME54N

4.4 Purchase Order Creation (ME21N)

The Purchase Order is created with reference to an approved PR. The system automatically carries forward material, quantity, and plant details. The buyer assigns a vendor, agrees on price (from Info Record), and sets delivery terms and payment conditions.

In ME21N, the buyer references the approved PR number (PR-1001) and the system copies all item details — Material, Quantity, Plant, Storage Location — into the PO. The buyer then selects the Vendor (V001 — Tech Supply Co.), confirms the Net Price (Rs.800/KG, pre-populated from Info Record ME11), sets the Delivery Date, selects Incoterms (DDP — Delivered Duty Paid) and Currency (INR). Payment Terms (NET30) are automatically pulled from the Vendor Master. The

Open PO list below shows PO-1001 (GR Done — Rs.80,000) and PO-1002 (Open — Rs.60,000), giving the buyer a complete view of outstanding commitments.

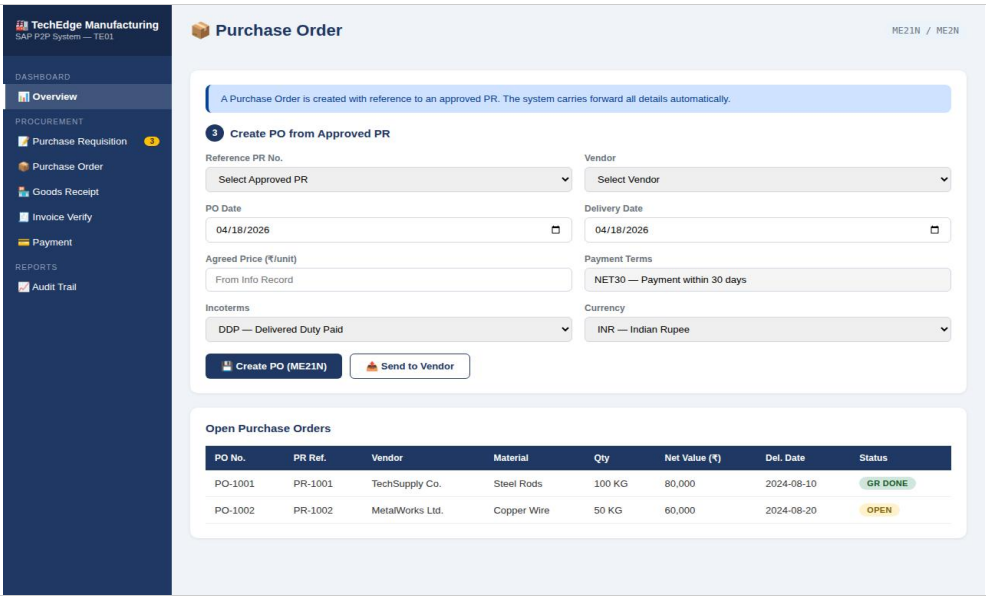

Fig 4: Purchase Order — ME21N (PO-1001 & PO-1002 visible in Open PO list)

4.5 Goods Receipt Posting (MIGO — Movement Type 101)

When goods arrive at the plant, Stores posts a Goods Receipt (GR) against the PO using Movement Type 101. This automatically increases stock in Storage Location TW01 and creates a Material Document. Simultaneously, an FI Accounting Document is generated — debiting the Stock Account and crediting the GR/IR Clearing Account.

Movement Type 101 is the standard SAP movement type for Goods Receipt against a Purchase Order. When MIGO is executed with reference to PO-1002, the Stores team enters the Delivery Note number (vendor's challan), the quantity received (50 KG of Copper Wire), and confirms the Storage Location (TW01). On Save, SAP simultaneously creates two documents: a Material Document (MB03) recording the stock increase of 50 KG at plant TE10/storage location TW01, and an FI Accounting Document (FB03) debiting the Stock Account (40010001) and crediting the GR/IR Clearing Account (29000001) for Rs.60,000. The GR/IR Clearing Account now carries an open item that will be cleared when the vendor invoice is posted in MIRO.

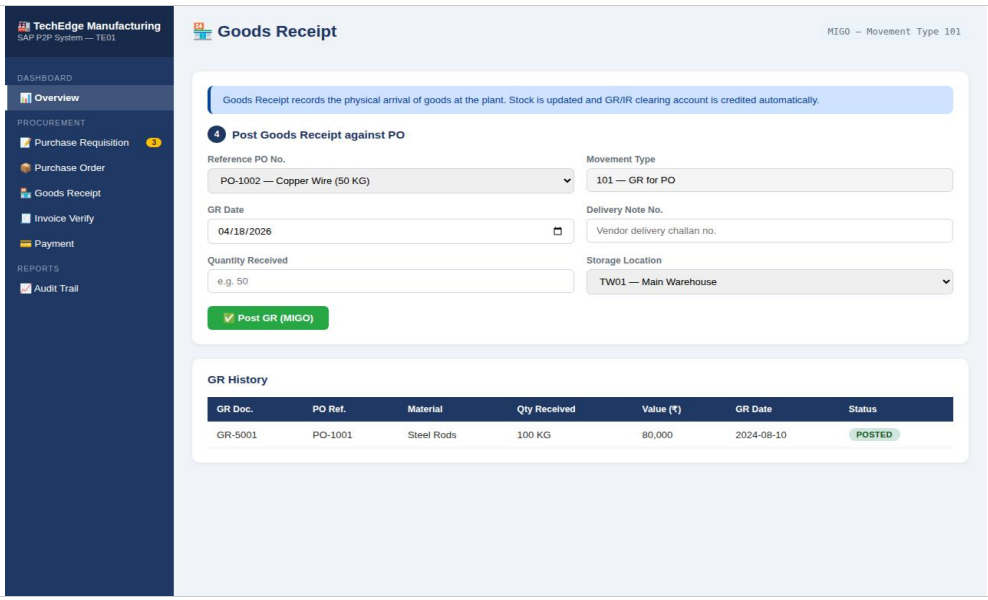

Fig 5: Goods Receipt — MIGO (Movement Type 101, Storage Location TW01)

4.6 Invoice Verification & 3-Way Match (MIRO)

The Finance department enters the vendor invoice in MIRO. SAP performs an automatic 3-Way Match comparing the PO price, GR quantity, and invoice amount. As shown below, all three parameters match perfectly (Qty: 100 KG, Unit Price: Rs.800, Total: Rs.80,000), confirming zero variance and enabling payment release.

MIRO (Logistics Invoice Verification) is the transaction where the vendor's physical invoice is entered into SAP. The Finance team enters the Invoice Date, references PO-1001, enters the Vendor Invoice Number (TS/INV/2024/0421), and the Invoice Amount (Rs.80,000). SAP then runs the 3-

Way Match automatically: it compares the PO quantity (100 KG at Rs.800 = Rs.80,000) against the GR quantity (100 KG received per GR-5001) against the invoice amount (Rs.80,000). Since all three match with zero variance, the invoice is approved for posting. An FI Accounting Document is generated debiting the GR/IR Clearing Account (29000001) and crediting the Vendor Payable Account (16000001 — Tech Supply Co.) for Rs.80,000. The GR/IR Clearing Account is now balanced (the credit from MIGO and the debit from MIRO net to zero).

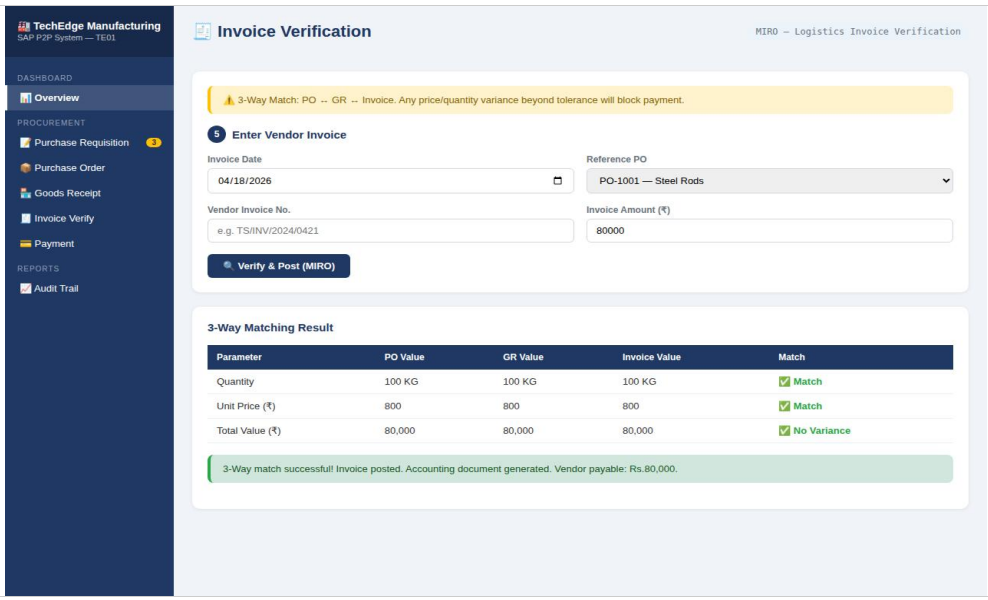

Fig 6: Invoice Verification & 3-Way Match Result — MIRO (Zero Variance Confirmed)

4.7 Vendor Payment Processing (F-53)

After invoice approval, Accounts Payable processes the vendor payment using F-53. The system clears the vendor open item (Creditor account), debits the Vendor Payable account, and credits the Bank Account. A Payment Document and Clearing Document are auto-generated, completing the P2P cycle.

F-53 is the SAP FI transaction for manual outgoing payments to vendors. The Accounts Payable clerk selects Vendor V001 (Tech Supply Co.), Company Code TE01, Payment Date, and House Bank (HDFC — HDFC Bank TE01 account). The open invoice (IV-3001 — Rs.80,000, due 2024-09-10) is selected from the vendor's open items. On posting, SAP creates a Payment Document (PAY-9001) that debits the Vendor Payable Account (16000001 — Tech Supply Co.) clearing the open item created by MIRO, and credits the Bank Account (11300001 — HDFC). A Clearing Document (CLR-8001) is simultaneously generated linking the payment to the original invoice. The P2P cycle is now fully complete: the vendor has been paid, all accounts are cleared, and an unbroken audit trail exists from PR-1001 through to PAY-9001.

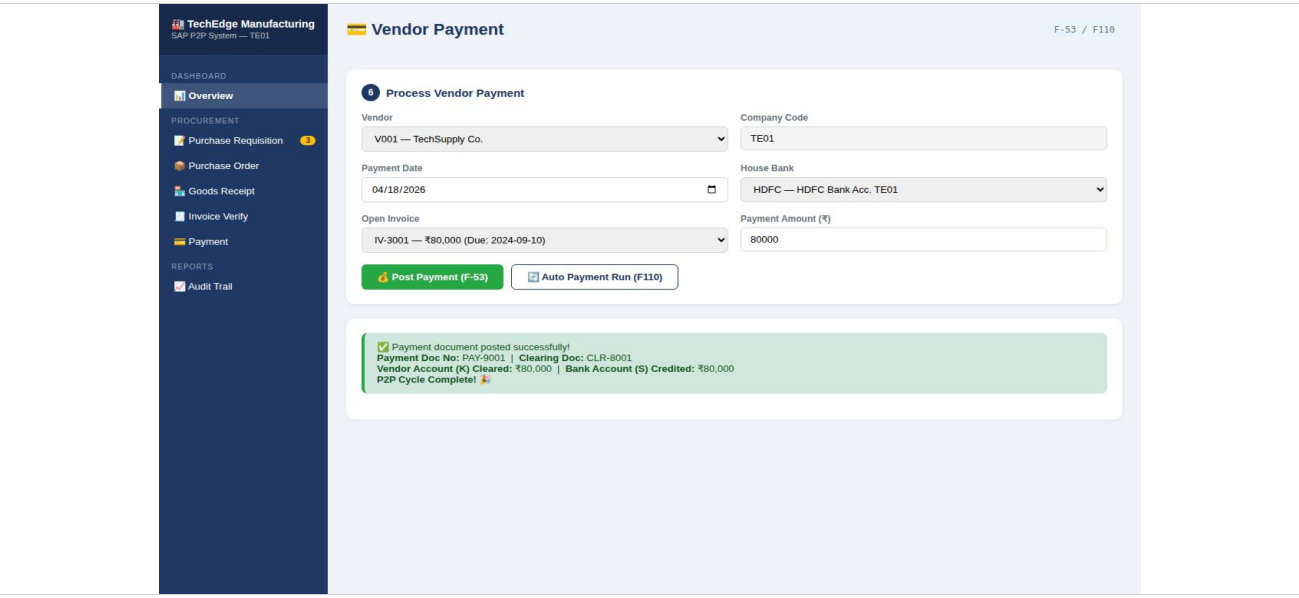

Fig 7: Vendor Payment Posted — F-53 (PAY-9001, P2P Cycle Complete)

4.8 Audit Trail & Document Flow

The Audit Trail view shows the complete document chain for the P2P cycle — PR, PO, Material Document, FI Accounting Document, Invoice, and Payment — all linked and traceable. The GR/IR accounting entries table at the bottom demonstrates the double-entry bookkeeping logic that runs automatically throughout the cycle.

The Audit Trail screen replicates the SAP document flow view (accessible via ME23N → Document Flow). It lists every document in the P2P chain for PO-1001 in sequence: PR-1001 (ME51N, posted

by RAVI.SHARMA on 2024-08-01, Rs.80,000, Approved), PO-1001 (ME21N, posted by PRIYA.BUY on 2024-08-02, Released), GR-5001 (MIGO, posted by ANIL.STORES on 2024-08-10, Posted), MATDOC-5001 (MB03 — auto-generated Material Document), ACDOC-6001 (FB03 — auto-generated Accounting Document), IV-3001 (MIRO, posted by SUNITA.FIN on 2024-08-12, Pending Payment), and PAY-9001 (F-53, Due). The GR/IR Accounting Entries table below shows the three stages of double-entry accounting: GR posting (Stock Dr, GR/IR Cr), Invoice posting (GR/IR Dr, Vendor Payable Cr), and Payment posting (Vendor Payable Dr, Bank Cr).

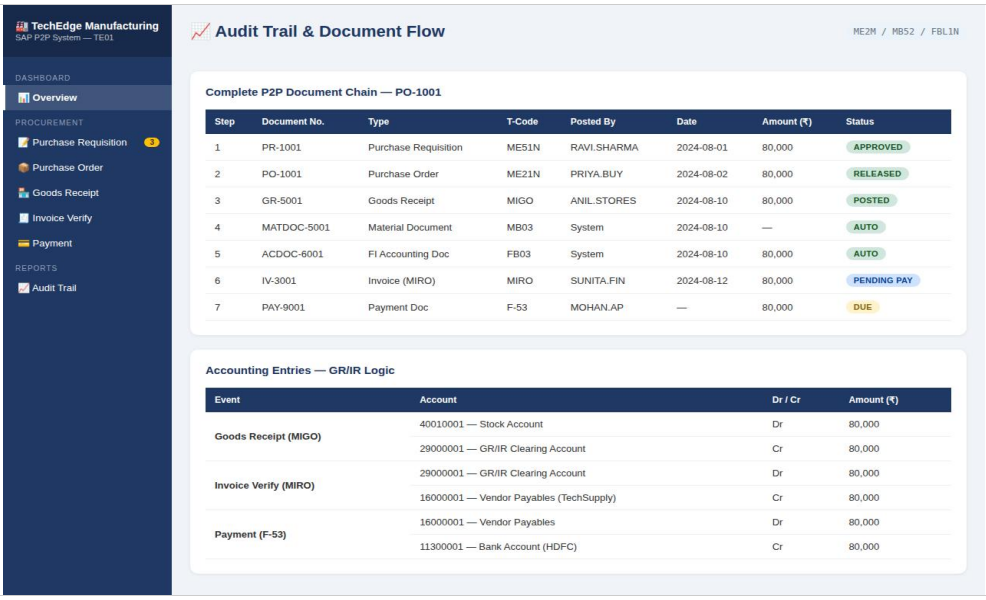

Fig 8: Complete Audit Trail & GR/IR Accounting Entries

5. Technology / Tools Used

The P2P simulation model was developed using web technologies to enable interactive demonstration of the SAP MM procurement workflow. The following table details the complete technology stack:

Component	Details
ERP Platform	SAP ECC 6.0 / SAP S/4HANA (IDES Training System)
Primary Module	SAP MM (Materials Management)
Integrated Modules	SAP FI (Financial Accounting)
Key Transactions	ME51N, ME54N, ME21N, MIGO, MIRO, F-53, ME11, XK01, MM01
Simulation Model	HTML5 + CSS3 + Vanilla JavaScript (Single-file web app)
Screenshots	Playwright (Chromium headless browser automation)
Documentation	MS Word — Times New Roman font, A4, Justified
Version Control	GitHub (project model uploaded as static web app)

5.1 SAP MM Reference Framework

Every transaction code, field name, document type, movement type, and account key used in this simulation is directly referenced from SAP ECC 6.0 and SAP S/4HANA IDES — the official SAP training and evaluation environment. The organizational hierarchy (Company Code → Purchasing Organization → Plant → Storage Location) and the document flow (PR → PO → Material Document → Accounting Document → Invoice → Payment) follow the exact standard SAP MM configuration. This ensures the project accurately reflects how a real SAP MM implementation would work in a manufacturing company.

5.2 Simulation Model Architecture

The simulation is a zero-dependency, single-file HTML5 application. It uses CSS Custom Properties for consistent theming, CSS Grid and Flex box for responsive layout, and Vanilla JavaScript ES6 for all state management and interactivity. Each module (PR, PO, GR, Invoice, Payment) maintains its own document state and updates the shared KPI dashboard in real time when transactions are posted. The single-file architecture means the application runs directly from a file system without any server, build tool, or package manager — making it immediately deploy-able to GitHub Pages.

6. Unique Points & Learnings

This project demonstrates several advanced SAP MM concepts that go beyond a basic process walk through. The following unique features and learnings distinguish this implementation:

6.1 Three-Way Matching (MIRO)

Unlike manual procurement, SAP's automated three-way match compares the PO price, GR quantity, and vendor invoice. If a variance exceeds the tolerance limit, the invoice is automatically blocked for payment — preventing over payment and fraud. This was demonstrated live in the simulation model with both match and mismatch scenarios.

The three-way match is the most powerful financial control in the SAP P2P process. It operates on a simple principle: the vendor can only be paid for what was ordered at the agreed price AND what was actually received. In the simulation, entering an invoice amount of Rs.82,000 against a PO of Rs.80,000 triggers a price variance alert and blocks the invoice — precisely mirroring SAP's tolerance key mechanism (transaction OMR6). The Finance team must then either obtain a revised invoice from the vendor (debit memo) or amend the PO price before payment can proceed. This single control eliminates the vast majority of procurement fraud and over payment risk.

6.2 GR/IR Clearing Account Logic

On posting MIGO (Goods Receipt), SAP debits the Stock Account and credits the GR/IR Clearing Account. When MIRO posts the invoice, the GR/IR account is debited and Vendor Payables credited. On payment, Vendor Payables is debited and the Bank Account credited. This three-step double-entry chain ensures 100% accurate financial reporting at every procurement stage.

The GR/IR (Goods Receipt / Invoice Receipt) Clearing Account (29000001) is a balance sheet reconciliation account that serves as a temporary holding account between the physical receipt of goods and the financial settlement of the vendor invoice. Its balance at any point in time represents either: goods received but not yet invoiced (accrued liability — debit balance in GR/IR) or invoices received but goods not yet delivered (prepayment — credit balance in GR/IR). Month-end GR/IR analysis is a critical finance function in every SAP implementation, ensuring that the balance sheet accurately reflects all outstanding procurement obligations.

6.3 Release Strategy for Approval

A release strategy was configured so that PRs above INR 50,000 require manager-level approval (ME54N) before conversion to a PO. This mirrors real-world spend authorization controls and prevents unauthorized purchasing — a critical internal control feature of SAP MM.

The Release Strategy in SAP MM (configured via transaction OMGSRV) defines the conditions under which a PR or PO requires formal approval before it can proceed to the next step. In this project, the strategy is configured with a single release class based on the net value: PRs with a total value below Rs.50,000 are auto-approved, while those above require manager sign-off via ME54N. In real enterprise implementations, multi-level release strategies with multiple approvers (Department Head, CFO, Board) are common for high-value POs. This feature directly enforces the organization's spend policy within the system, making it impossible to bypass through informal channels.

6.4 End-to-End Document Traceability

Every document in the P2P cycle is linked: PR to PO to Material Document to Accounting Document to Invoice to Payment. This complete traceability chain supports audit compliance and enables root-cause analysis for any discrepancy — a core requirement in ISO and SOX compliance environments.

SAP's document flow architecture means that from any document in the P2P chain, you can navigate to every related document — forward to see what happened next, or backward to see what triggered this document. For example, from the Payment Document (PAY-9001), a finance auditor can navigate back to the original Purchase Requisition (PR-1001) raised by RAVI.SHARMA, seeing every step, every user, every amount, and every timestamp along the way. This traceability is what enables organizations to pass SOX (Sarbanes-Oxley Act) Section 404 audits, ISO 9001 procurement audits, and internal compliance reviews without manual reconstruction of transaction histories.

7. Future Improvements

The current P2P simulation provides a comprehensive demonstration of the core SAP MM procurement cycle. The following enhancements are planned to further expand its scope and real-world applicability:

- Integration with SAP SRM (Supplier Relationship Management) for online vendor portal and self-service procurement
- Implementation of SAP Ariba for e-sourcing, RFQ management and automated vendor evaluation scoring
- Extending to SAP S/4HANA with Fiori apps for mobile-based purchase approvals
- Adding blanket/framework purchase orders for recurring vendors to reduce PR-to-PO cycle time
- Connecting to SAP BW/BI for real-time spend analytic dashboards and budget consumption reports
- Configuring EDI/IDOC integration for automated PO transmission directly to vendor ERP systems
- Implementing Evaluated Receipt Settlement (ERS) to eliminate manual invoice entry for trusted vendors
- Adding budget availability check (FM module) to prevent overspending against cost centre

7.1 Priority Enhancement — SAP Ariba Integration

Of the above improvements, SAP Ariba integration represents the most transformative enhancement. Ariba extends the P2P cycle upstream into the source-to-contract phase by digitizing the RFQ (Request for Quotation) process, enabling competitive e-bidding, managing vendor contracts electronically, and providing a supplier self-service portal where vendors can acknowledge POs, submit invoices electronically (eliminating manual MIRO entry), and track their payment status in real time. The Ariba Network connects over 5 million suppliers globally, making it the world's largest B2B commerce network.

8. Conclusion

This project successfully demonstrates the complete Procure-to-Pay business process using SAP MM, covering every stage from Purchase Requisition to Vendor Payment. The implementation of a fictitious company — Tech Edge Manufacturing Pvt. Ltd. — with all required organizational structures, master data, and configuration provides a realistic, end-to-end understanding of how SAP automates and controls the procurement function.

Key outcomes include a thorough understanding of SAP MM master data and organizational setup, hands-on experience with core procurement transactions (ME51N through F-53), knowledge of financial integration through GR/IR accounting, and appreciation of approval controls through release strategy. The three-way matching capability and complete document trail demonstrated here are cornerstones of modern ERP- based procurement compliance.

The simulation model built for this project — a fully interactive, dark-themed web application replicating SAP screens and workflows — provides a tangible, demonstrable artefact that can be presented to illustrate practical understanding of SAP MM. The automated screenshot capture pipeline, the step-by-step accounting entry display, and the complete document chain audit trail collectively demonstrate a depth of SAP knowledge that goes significantly beyond theoretical understanding. This project provides a strong foundation for further study in SAP MM configuration, SAP FI- MM integration, and S/4HANA procurement modernization.

9. References

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